



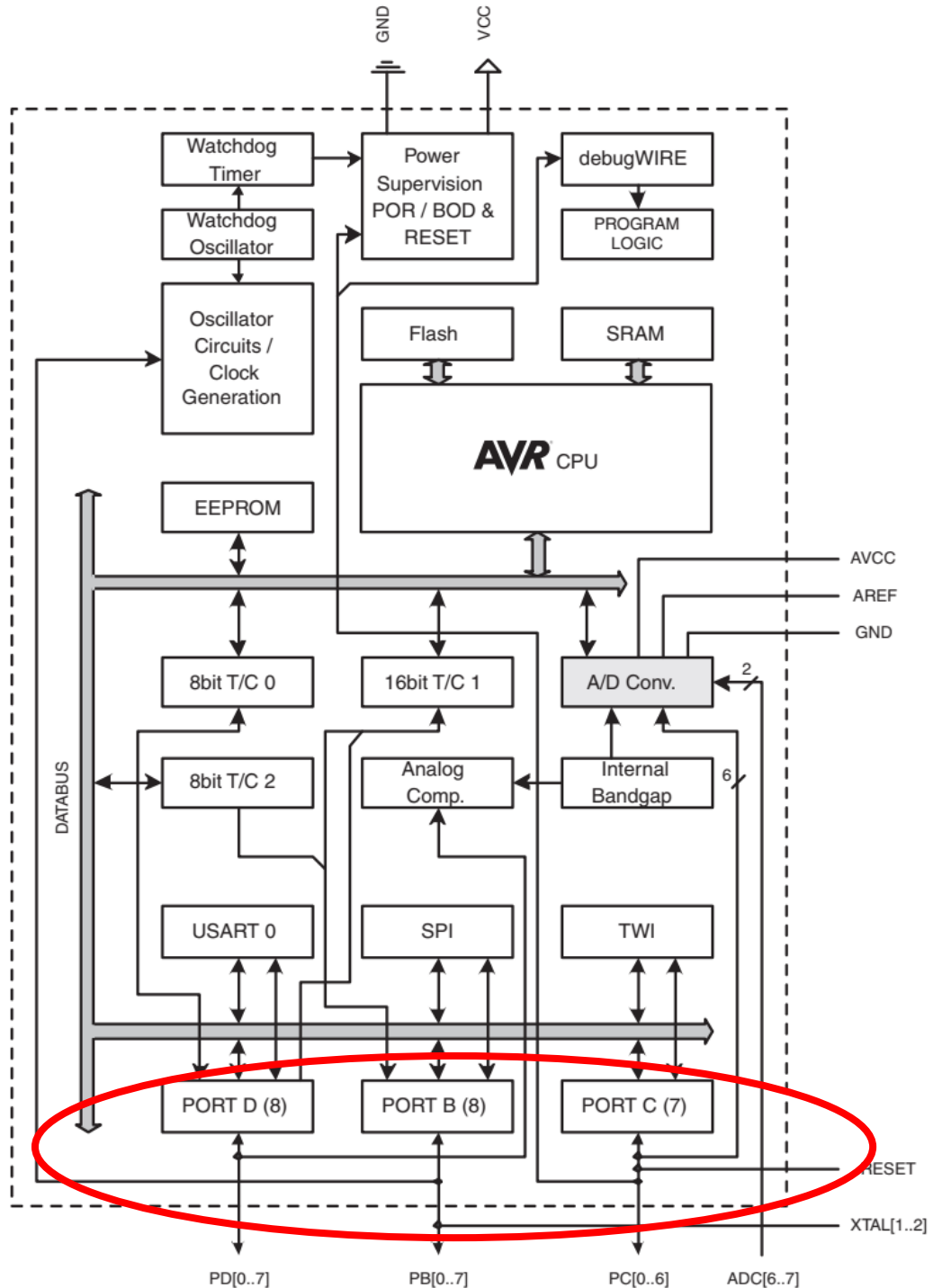
# CIRCUITOS DIGITALES Y MICROCONTROLADORES 2025

Facultad de Ingeniería  
UNLP

Puertos de entrada salida  
Características eléctricas

Ing. José Juárez

# AVR - ATMEGA 328P



# Registros de los Puertos I/O

- Ejemplo puerto B

## PORTB – Port B Data Register Address I/O: 0x05

| Bit           | 7      | 6      | 5      | 4      | 3      | 2      | 1      | 0      |       |
|---------------|--------|--------|--------|--------|--------|--------|--------|--------|-------|
|               | PORTB7 | PORTB6 | PORTB5 | PORTB4 | PORTB3 | PORTB2 | PORTB1 | PORTB0 | PORTB |
| Read/Write    | R/W    | R/W    | R/W    | R/W    | R/W    | R/W    | R/W    | R/W    |       |
| Initial Value | 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      |       |

## DDRB – Port B Data Direction Register Address I/O: 0x04

| Bit           | 7    | 6    | 5    | 4    | 3    | 2    | 1    | 0    |      |
|---------------|------|------|------|------|------|------|------|------|------|
|               | DDB7 | DDB6 | DDB5 | DDB4 | DDB3 | DDB2 | DDB1 | DDB0 | DDRB |
| Read/Write    | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  | R/W  |      |
| Initial Value | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |      |

## PINB – Port B Input Pins Address Address I/O: 0x03

| Bit           | 7     | 6     | 5     | 4     | 3     | 2     | 1     | 0     |      |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|               | PINB7 | PINB6 | PINB5 | PINB4 | PINB3 | PINB2 | PINB1 | PINB0 | PINB |
| Read/Write    | R     | R     | R     | R     | R     | R     | R     | R     |      |
| Initial Value | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |      |

descripción

mundo  
lógico

# Registros de los Puertos I/O

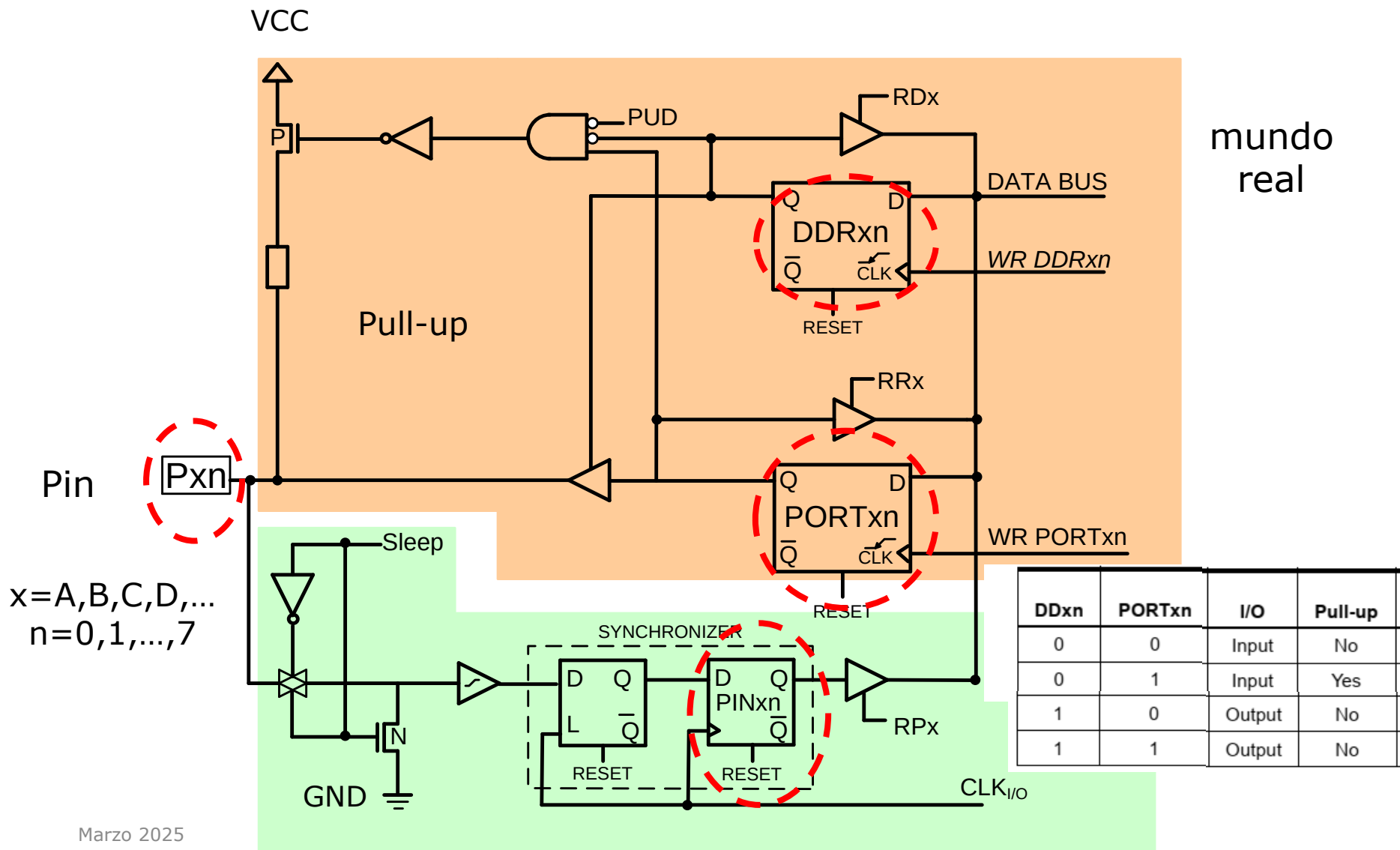
- Como se configura la dirección

| DDxn | PORTxn | I/O    | Pull-up | Comment                                     |
|------|--------|--------|---------|---------------------------------------------|
| 0    | 0      | Input  | No      | Tri-state (Hi-Z)                            |
| 0    | 1      | Input  | Yes     | Pxn will source current if ext. pulled low. |
| 1    | 0      | Output | No      | Output Low (Sink)                           |
| 1    | 1      | Output | No      | Output High (Source)                        |

Ver ejemplo de simulación del TP1

# Estructura de los Puertos I/O

*Atención: circuito completo en pág. 85 del data sheet*



# Estructura de los Puertos I/O

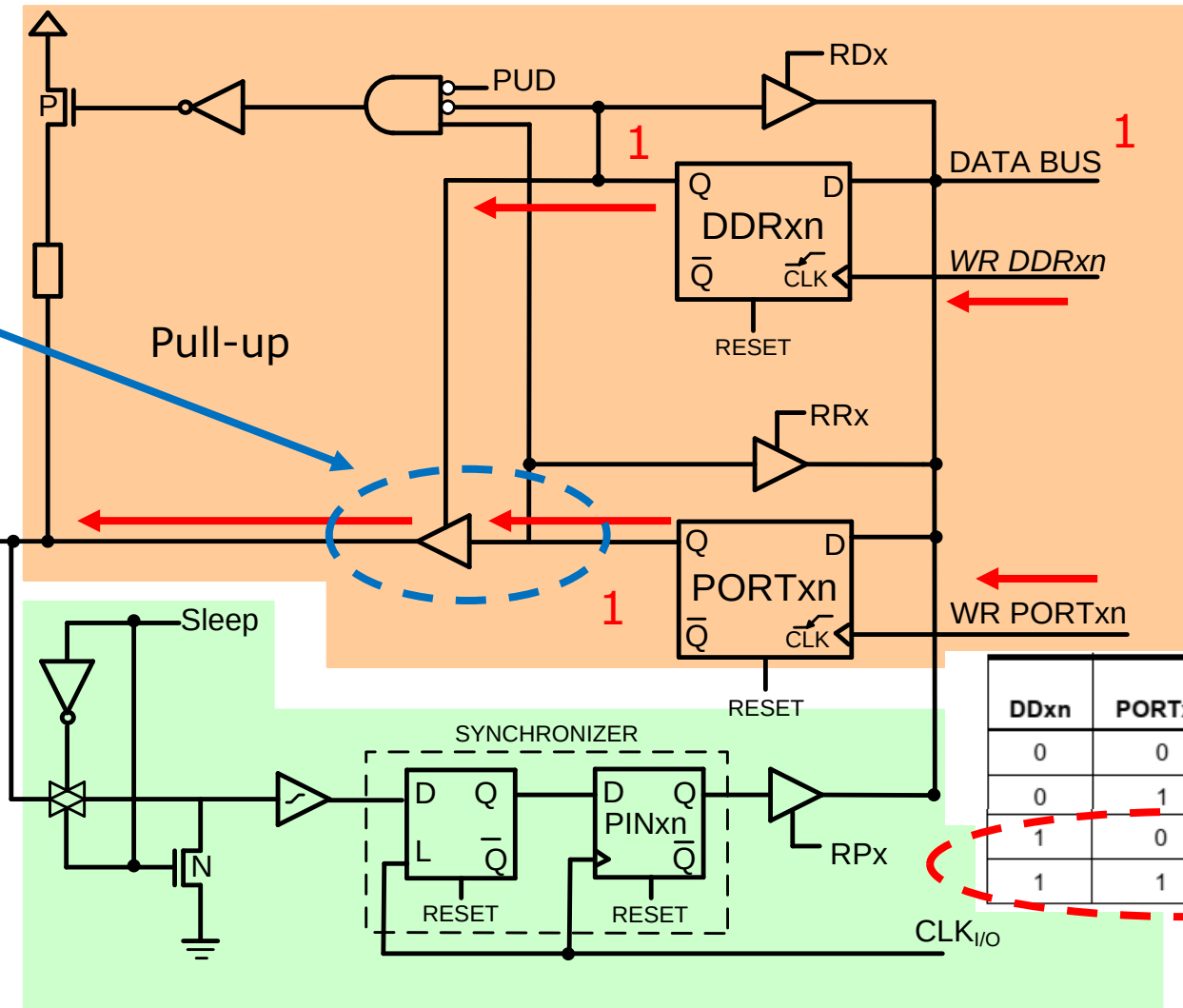
Repasar de la  
bibliografía  
Introducción al  
Diseño Lógico

Driver de salida de 3 estados

Pin 1 Pxn

$$x=A,B,C,D,\dots$$
$$n=0,1,\dots,7$$

como  
salida

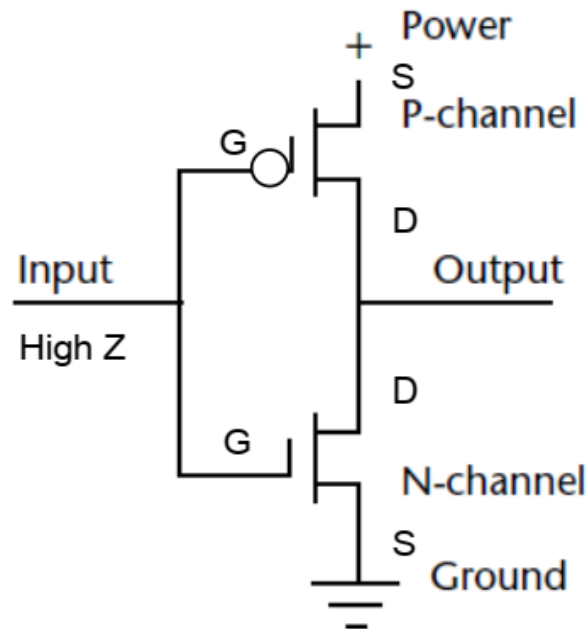
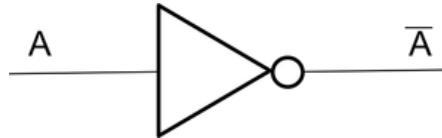


| DDxn | PORTxn | I/O    | Pull-up |
|------|--------|--------|---------|
| 0    | 0      | Input  | No      |
| 0    | 1      | Input  | Yes     |
| 1    | 0      | Output | No      |
| 1    | 1      | Output | No      |

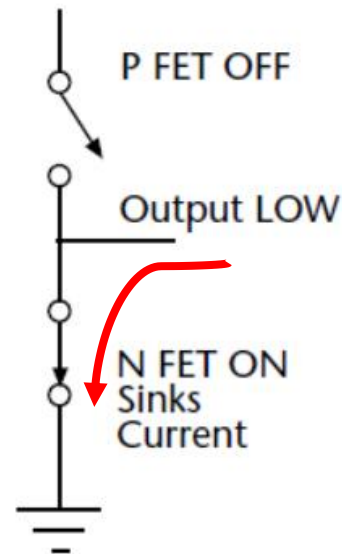
# Introducción a la lógica CMOS

- Para comprender las características eléctricas de los puertos I/O repasemos...

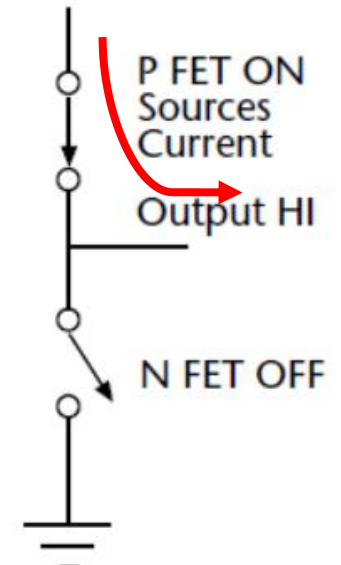
- Inversor CMOS IDEAL



Output LOW



Output HI



- CMOS(Complementary MOSFET)
- MOSFET(Metal Oxide Semiconductor Field Effect Transistor)

Marzo 2025

Repasar CMOS de la bibliografía:  
*Sistemas Digitales: Principios y aplicaciones*, R. Tocci el CH8

# Introducción a la lógica CMOS

Por convención:  
el signo de  $I_{OH}$  se toma (-)  
el signo de  $I_{IH}$  se toma (+)

- Conexión entre dos dispositivos CMOS y sentido de la corriente

Capacidad de corriente de una salida  
(Fan-Out):

Dependerá del Tipo y N° de compuertas  
conectadas a esa salida.

características DC y AC de las compuertas:

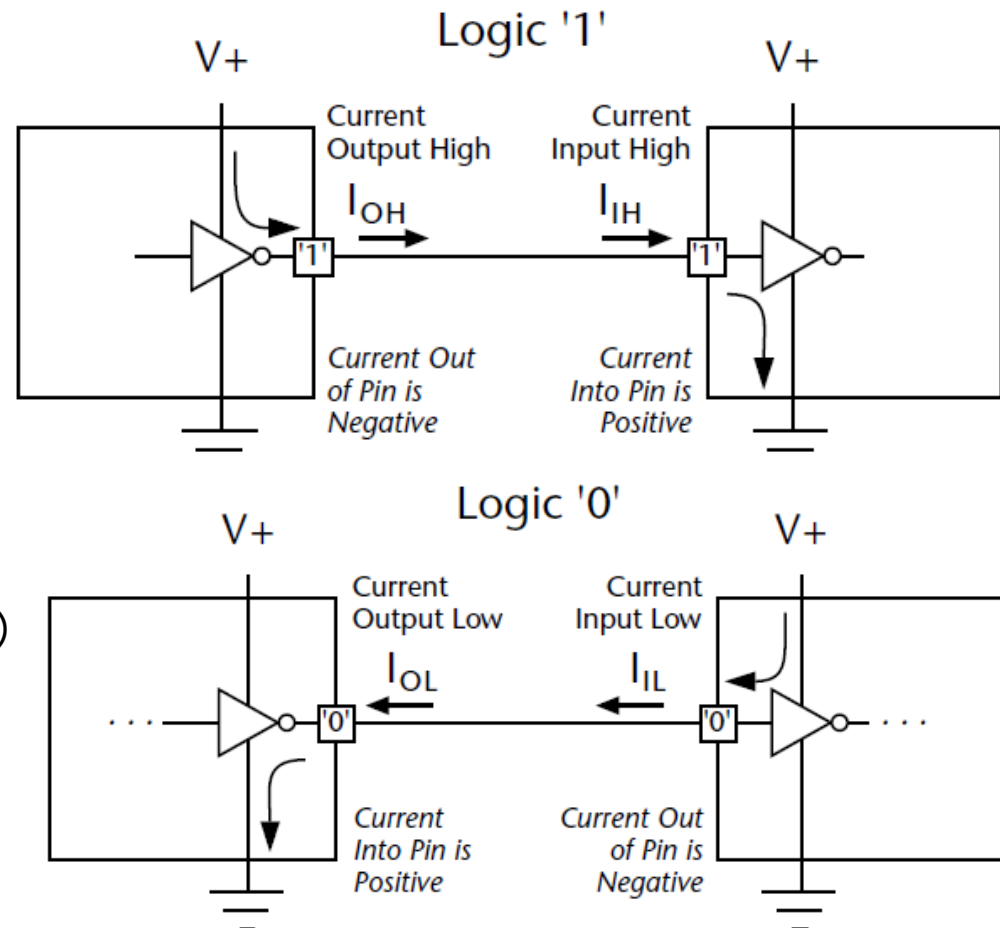
**Características DC:**

- $V_{OLmax}$ ,  $I_{OHmax}$ ,  $I_{OLmax}$ ,  $I_{OHmax}$
- $V_{ILmax}$ ,  $V_{IHmax}$ ,  $I_{ILmax}$ ,  $I_{IHmax}$

**Características AC:**

- $C_L$ : capacidad de carga a la salida (test)
- $C_{in}$ : capacidad de entrada
- $C_{stray}$ : Capacidad del cableado y PCB

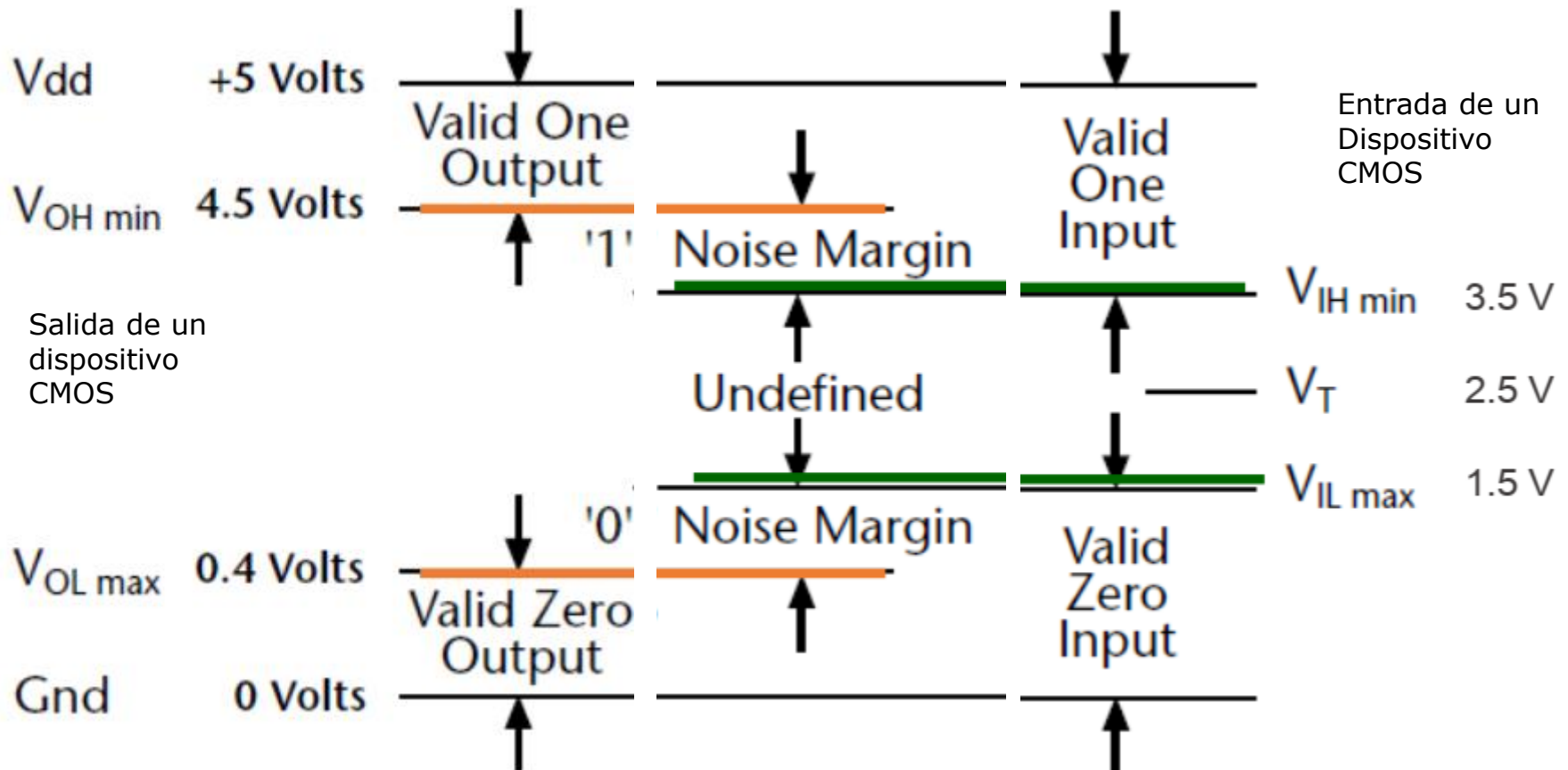
$$C_L > C_{in1} + C_{in2} + \dots + C_{stray}$$





# Introducción a la lógica CMOS

- Valores de operación y margen de ruido (características DC) típicos en familias CMOS de 5V



# Especificación para AVR

- Valores de referencia MCU ATMEGA328P:

## 28.2 DC Characteristics

$T_A = -40^\circ\text{C}$  to  $85^\circ\text{C}$ ,  $V_{CC} = 1.8\text{V}$  to  $5.5\text{V}$  (unless otherwise noted)

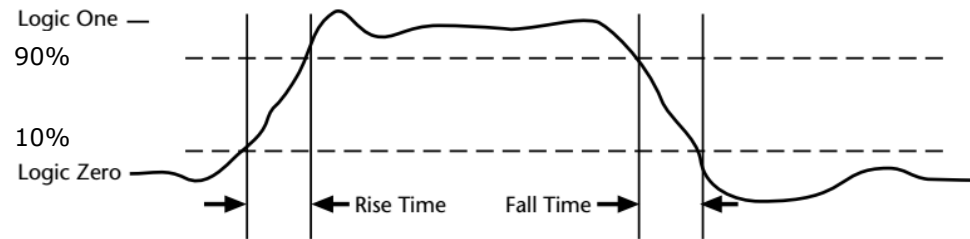
| Symbol   | Parameter                                              | Condition                                                                                          | Min.                                   | Typ. | Max.                                   | Units |
|----------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------|------|----------------------------------------|-------|
| $V_{IL}$ | Input Low Voltage, except XTAL1 and RESET pin          | $V_{CC} = 1.8\text{V} - 2.4\text{V}$<br>$V_{CC} = 2.4\text{V} - 5.5\text{V}$                       | -0.5<br>-0.5                           |      | $0.2V_{CC}^{(1)}$<br>$0.3V_{CC}^{(1)}$ | V     |
| $V_{IH}$ | Input High Voltage, except XTAL1 and RESET pins        | $V_{CC} = 1.8\text{V} - 2.4\text{V}$<br>$V_{CC} = 2.4\text{V} - 5.5\text{V}$                       | $0.7V_{CC}^{(2)}$<br>$0.6V_{CC}^{(2)}$ |      | $V_{CC} + 0.5$<br>$V_{CC} + 0.5$       | V     |
| $V_{OL}$ | Output Low Voltage <sup>(3)</sup><br>except RESET pin  | $I_{OL} = 20\text{ mA}$ , $V_{CC} = 5\text{V}$<br>$I_{OL} = 10\text{ mA}$ , $V_{CC} = 3\text{V}$   |                                        |      | 0.9<br>0.6                             | V     |
| $V_{OH}$ | Output High Voltage <sup>(4)</sup><br>except Reset pin | $I_{OH} = -20\text{ mA}$ , $V_{CC} = 5\text{V}$<br>$I_{OH} = -10\text{ mA}$ , $V_{CC} = 3\text{V}$ | 4.2<br>2.5                             |      |                                        | V     |

Prestar atención que depende si uso alimentación Vcc de 3V o de 5V y de la corriente de salida que esté requiriendo el circuito

# Introducción a la lógica CMOS

- Temporización (características AC)

Tiempo de Subida  
Tiempo de Bajada



Retardo de Propagación

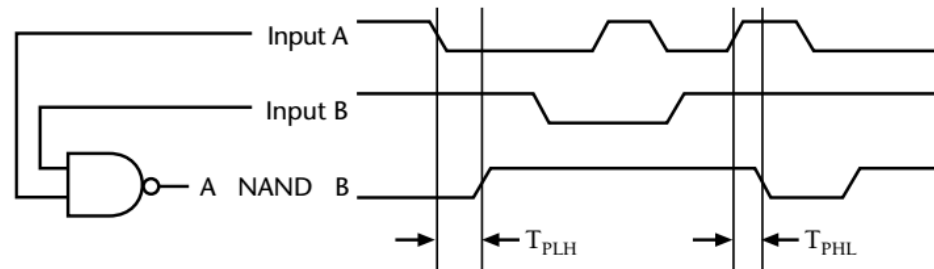
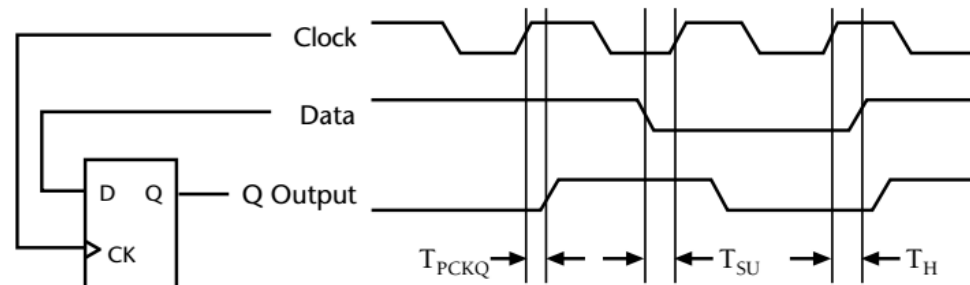


Figure 3-3: Propagation delay.

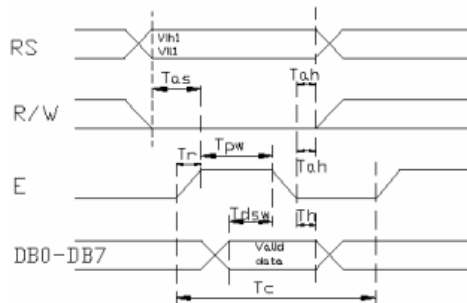
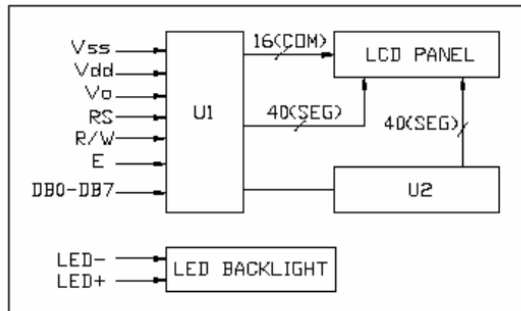
Tiempo de Setup  
Tiempo de Hold



# Introducción a la lógica CMOS

De la hoja de datos del Display  
WH1602A-YGH-CTK.pdf  
Familia TTL

- Ejemplo: quiero conectar mi MCU con este dispositivo digital:



## 5. DC CHARACTERISTICS (Ta=25°C; Vdd=5.0V±5%, Vss=0V)

| Item                                      | Symbol | Test Condition | Min.    | Typ. | Max. | Unit |
|-------------------------------------------|--------|----------------|---------|------|------|------|
| Operating Voltage                         | Vdd    | --             | --      | 5.0  | --   | V    |
| LCD Voltage                               | Vlcd   | Vdd-Vo         | --      | 5.0  | --   | V    |
| Operating Current                         | Idd    | Vdd=5.0V       | --      | 1.2  | 1.8  | MA   |
| Input "High" Voltage (1)<br>(Except OSC1) | Vih1   | --             | 2.2     | --   | Vdd  | V    |
| Input "Low" Voltage (1)<br>(Except OSC1)  | Vil1   | --             | -0.3    | --   | 0.6  | V    |
| Input "High" Voltage (2)<br>(OSC1)        | Vih2   | --             | Vdd-1.0 | --   | Vdd  | V    |
| Input "Low" Voltage (2)<br>(OSC1)         | Vil2   | --             | -0.2    | --   | 1.0  | V    |
| Output "High" Voltage (1)<br>(D0-D7)      | Voh1   | Ioh=-0.205mA   | 2.4     | --   | --   | V    |
| Output "Low" Voltage (1)<br>(D0-D7)       | Vol1   | Iol=1.2mA      | --      | --   | 0.4  | V    |

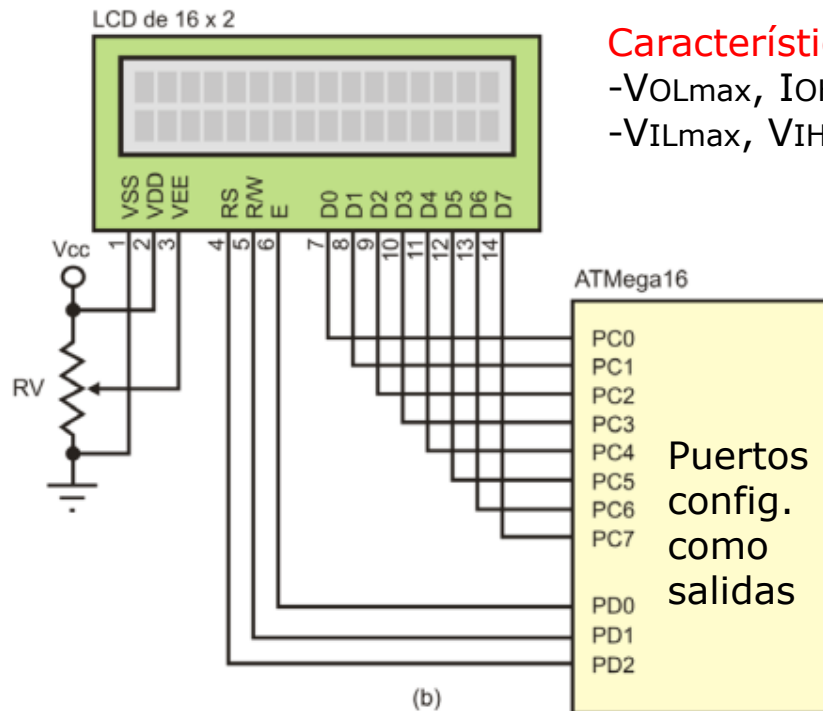
## 8. AC CHARACTERISTICS (Vdd=5.0V±5%, Vss=0V, Ta=25°C)

(Write mode)

| Characteristic            | Symbol | Min. | Typ. | Max. | Unit |
|---------------------------|--------|------|------|------|------|
| E Cycle Time              | Tc     | 500  | --   | --   | ns   |
| E Rise/Fall Time          | Tr,Tf  | --   | --   | 25   | ns   |
| E Pulse Width (High, Low) | Tpw    | 220  | --   | --   | ns   |
| R/W and RS Set-up Time    | Tas    | 40   | --   | --   | ns   |
| R/W and RS Hold Time      | Tah    | 10   | --   | --   | ns   |
| Data Set-up Time          | Tdsw   | 60   | --   | --   | ns   |
| Data Hold Time            | Th     | 10   | --   | --   | ns   |

# Introducción a la lógica CMOS

- Solución:



## Características DC:

- $V_{OLmax}$ ,  $I_{OHmax}$ ,  $I_{OLmax}$ ,  $I_{OHmax}$
- $V_{ILmax}$ ,  $V_{IHmax}$ ,  $I_{ILmax}$ ,  $I_{IHmax}$

Repasar interconexión de familias lógicas del libro de Tocci el CH8

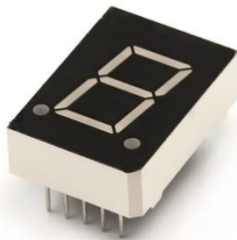
*¿Son eléctricamente compatibles?*

# Introducción a la lógica CMOS

- Ahora quiero conectar mi MCU con estos periféricos



*¿Son eléctricamente compatibles?*

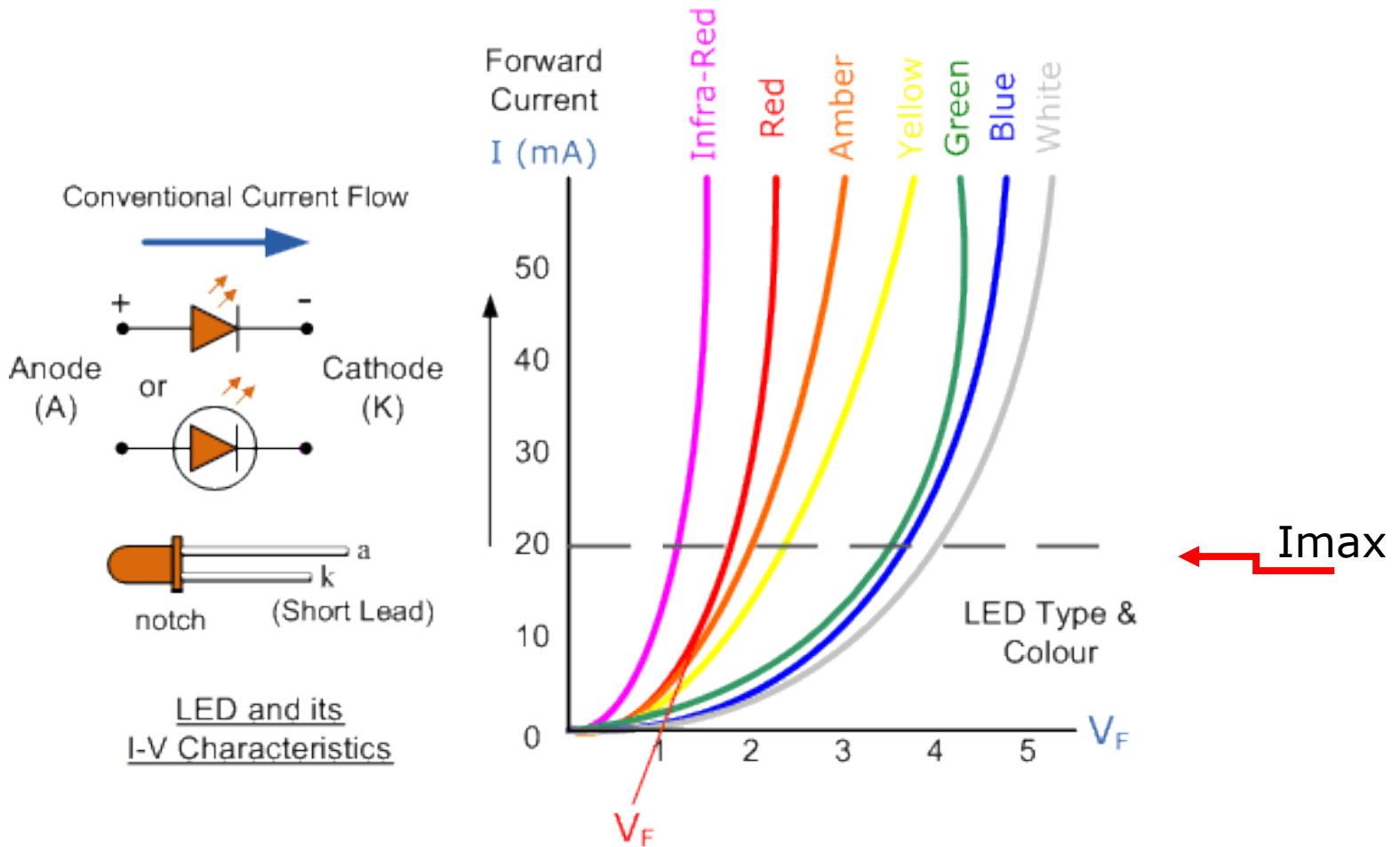


## Características DC:

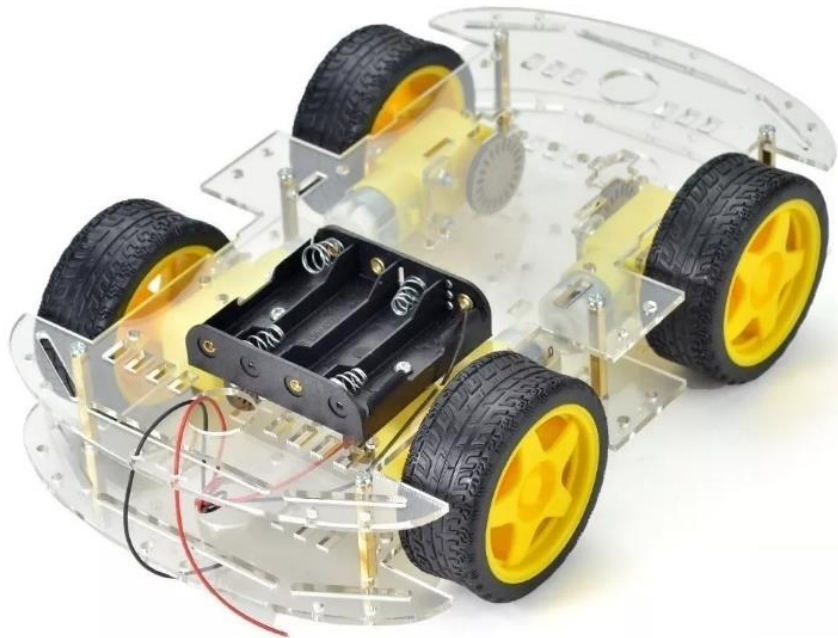
- VOLmax, IOHmax, IOLmax, IOHmax
- VILmax, VIHmax, IILmax, IIHmax



# Diodos LED (Light Emitting Diode)



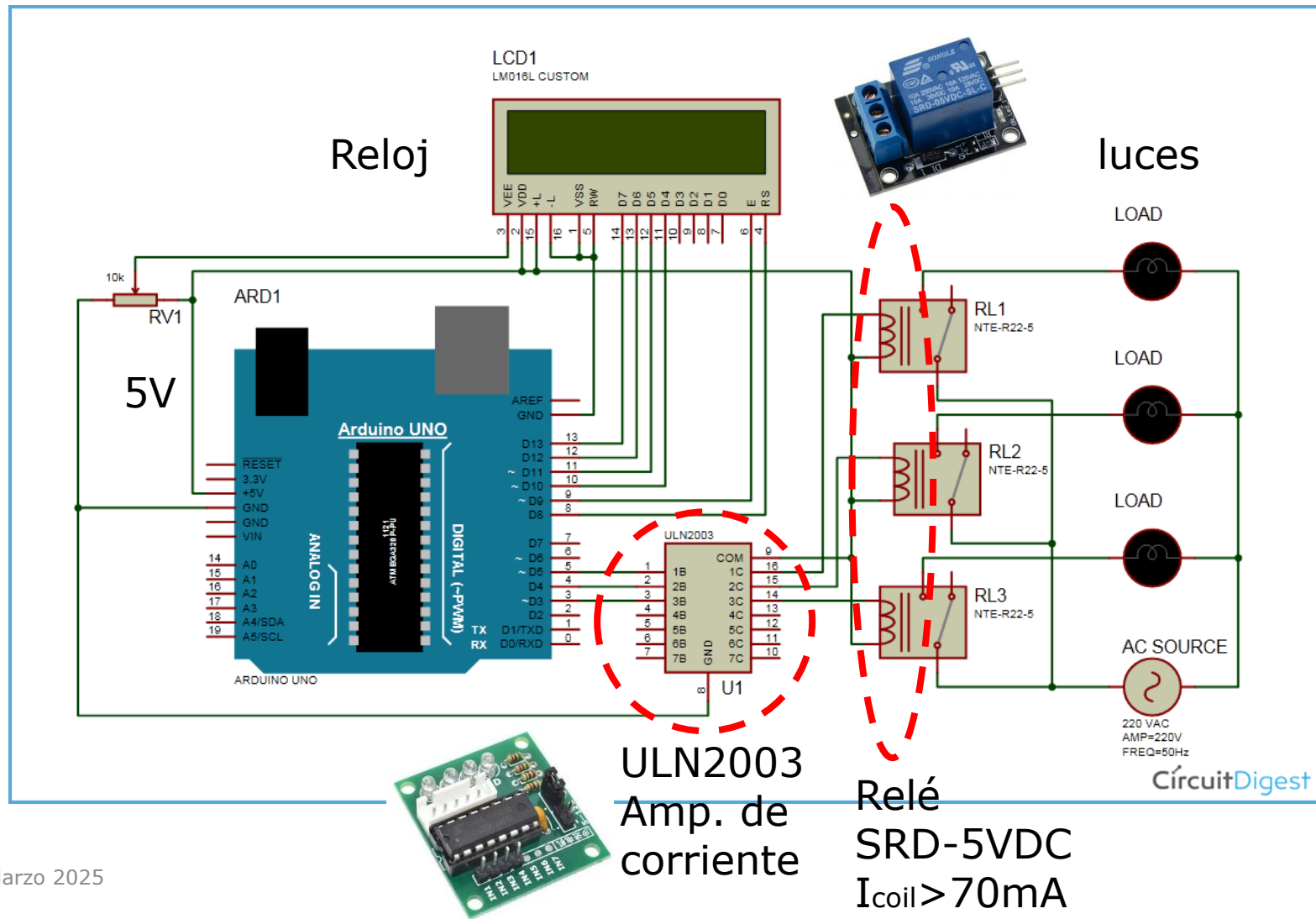
# Motor DC con reducción y rueda



Voltage: 3-6V  
Consumo: 80-150mA  
velocidad: 95-200 RPM

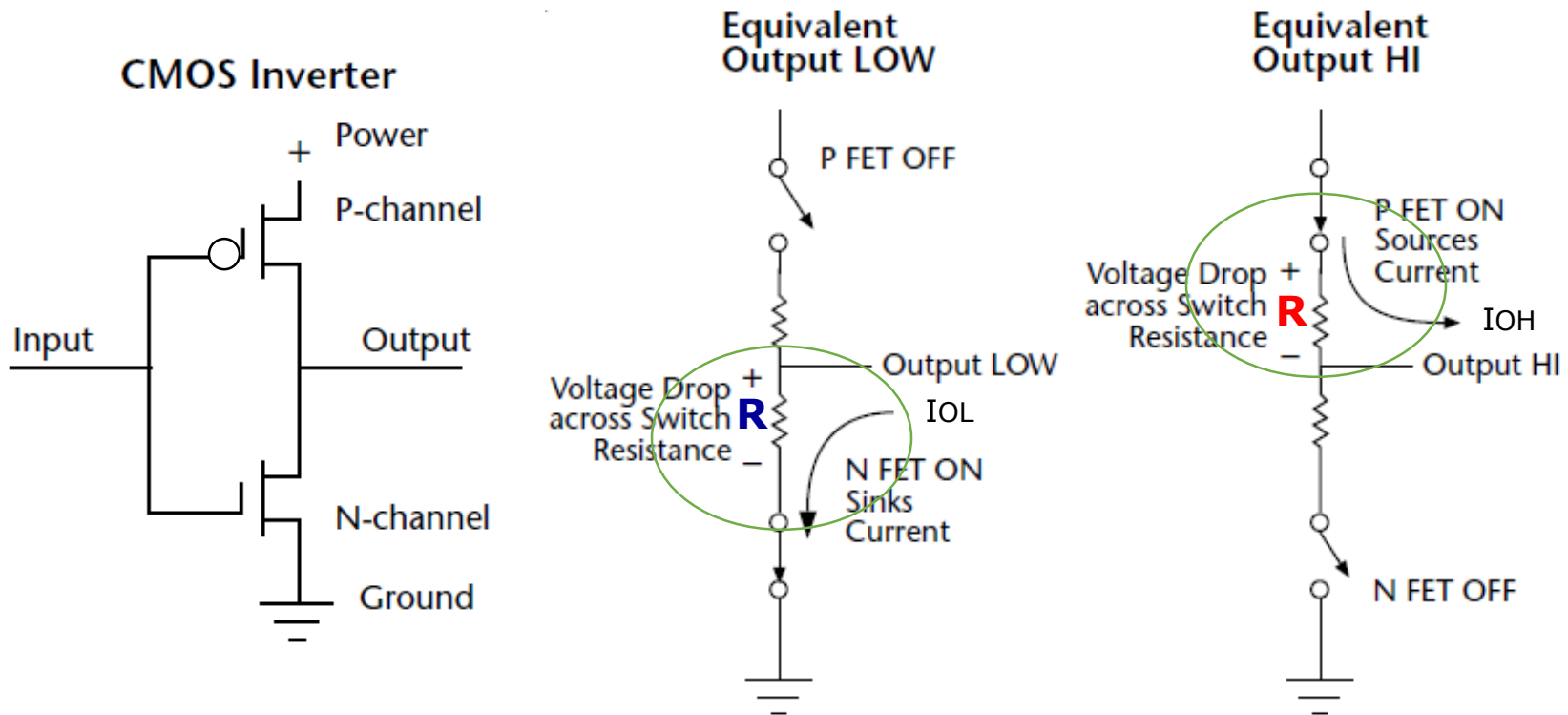


# Otro ejemplo: domótica Fácil



# Características eléctricas de salidas CMOS

- ¿Puedo conectar al puerto del MCU directamente un LED, un motor DC, un relé? veamos...
- Inversor CMOS real



$$V_{OL} = I_{OL} \cdot R$$

$$P_{OL} = (I_{OL})^2 \cdot R \text{ [W]}$$

$$V_{OH} = V_{CC} - I_{OH} \cdot R$$

$$P_{OH} = (I_{OH})^2 \cdot R \text{ [W]}$$

# Especificación para el ATmega328p

- Corriente de salida vs Tensión de Salida: nivel alto "1"

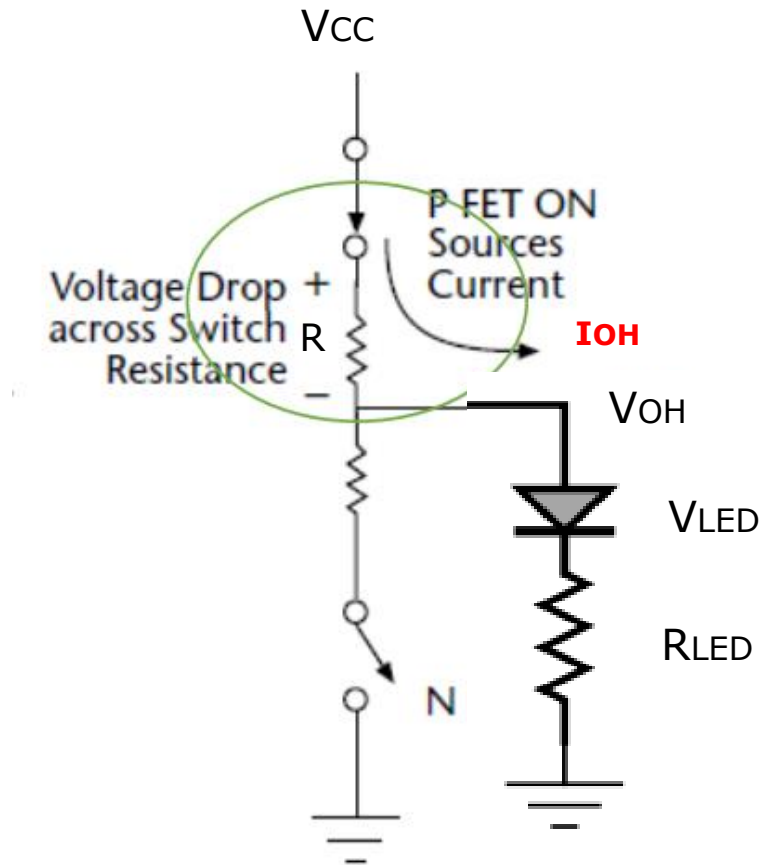
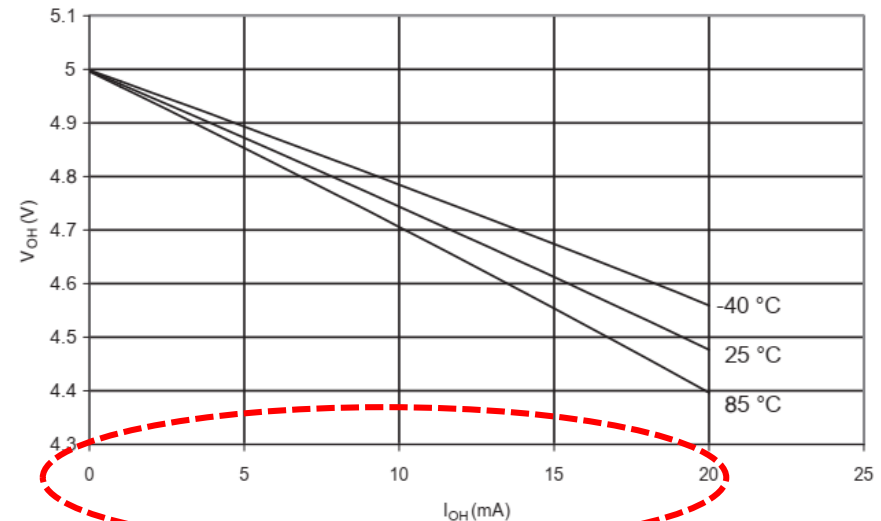


Figure 29-162. ATmega328P: I/O Pin Output Voltage vs. Source Current ( $V_{CC} = 5\text{ V}$ )



Zonas seguras de operación

$$V_{OH} = V_{DD} - I_{OH} \cdot R$$

$$I_{OH} = (V_{OH} - V_{LED}) / R_{LED}$$

# Especificación para el ATmega328p

- Corriente de salida vs Tensión de Salida: nivel bajo "0"

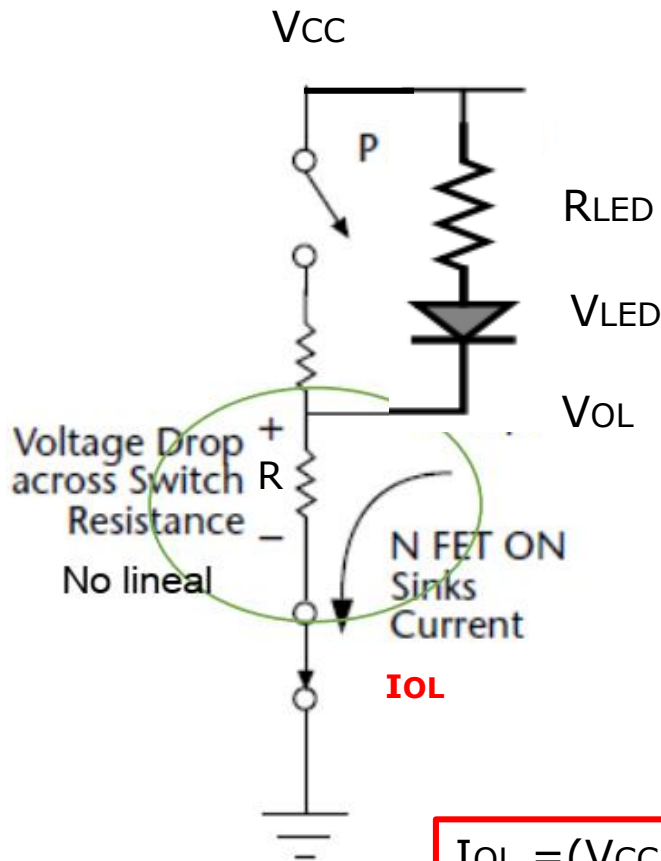
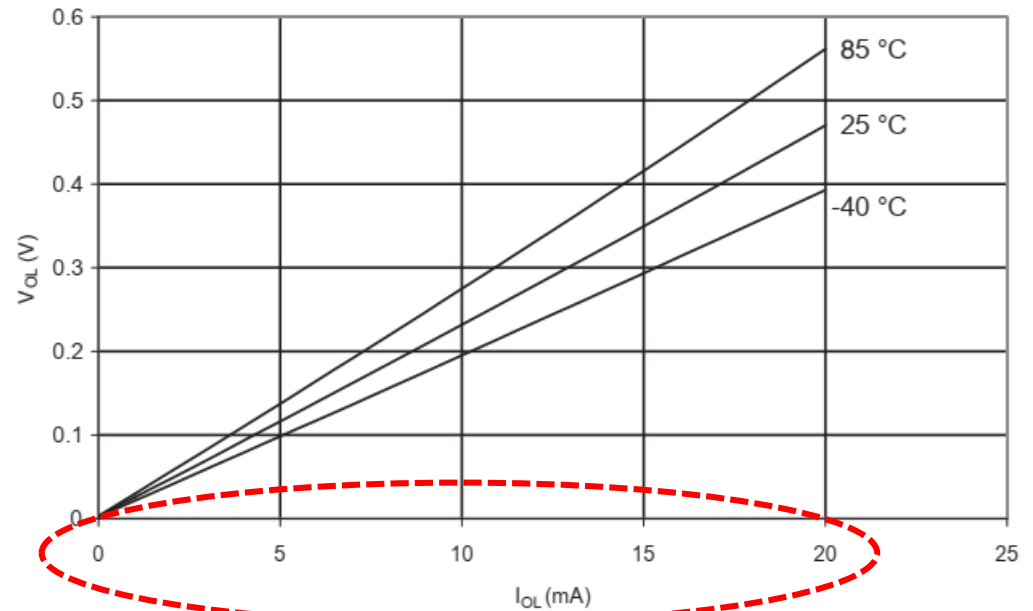


Figure 29-160. ATmega328P: I/O Pin Output Voltage vs. Sink Current ( $V_{CC} = 5\text{ V}$ )



Zonas seguras de operación

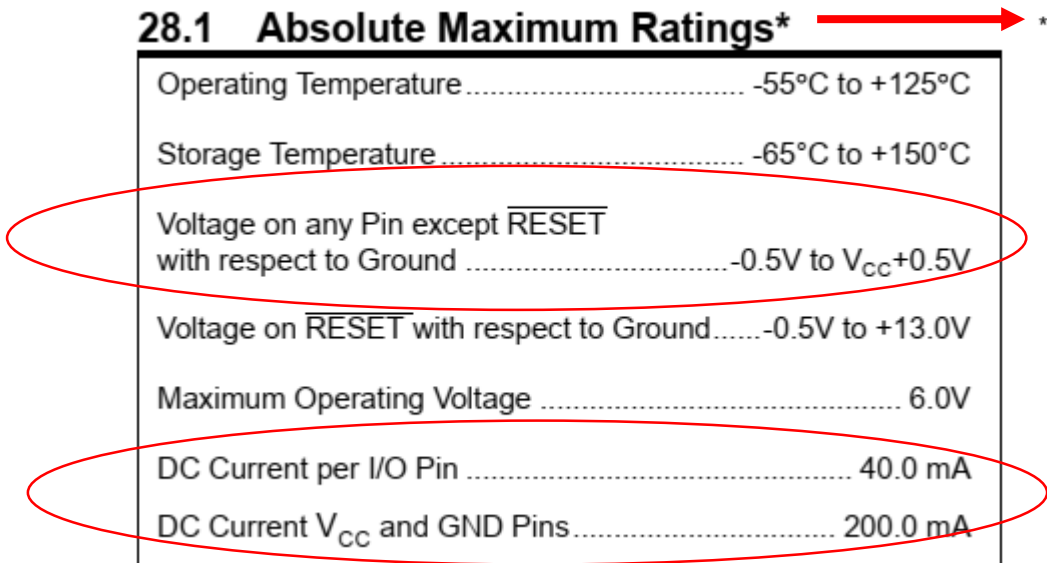
$$I_{OL} = (V_{CC} - V_{OL} - V_{LED}) / R_{LED}$$

$$V_{OL} = I_{OH} \cdot R$$

# Especificación para el ATmega328p

- Condiciones Máximas para Operación:

## 28.1 Absolute Maximum Ratings\*



|                                                                                     |                        |
|-------------------------------------------------------------------------------------|------------------------|
| Operating Temperature .....                                                         | -55°C to +125°C        |
| Storage Temperature .....                                                           | -65°C to +150°C        |
| Voltage on any Pin except $\overline{\text{RESET}}$<br>with respect to Ground ..... | -0.5V to $V_{CC}+0.5V$ |
| Voltage on $\overline{\text{RESET}}$ with respect to Ground .....                   | -0.5V to +13.0V        |
| Maximum Operating Voltage .....                                                     | 6.0V                   |
| DC Current per I/O Pin .....                                                        | 40.0 mA                |
| DC Current $V_{CC}$ and GND Pins .....                                              | 200.0 mA               |

\*NOTICE:

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or other conditions beyond those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

# Especificación para el ATmega328p

- Condiciones Máximas para Operación:

|          |                                                        |                                                                                  |            |  |            |   |
|----------|--------------------------------------------------------|----------------------------------------------------------------------------------|------------|--|------------|---|
| $V_{OL}$ | Output Low Voltage <sup>(3)</sup><br>except RESET pin  | $I_{OL} = 20 \text{ mA}, V_{CC} = 5V$<br>$I_{OL} = 10 \text{ mA}, V_{CC} = 3V$   |            |  | 0.9<br>0.6 | V |
| $V_{OH}$ | Output High Voltage <sup>(4)</sup><br>except Reset pin | $I_{OH} = -20 \text{ mA}, V_{CC} = 5V$<br>$I_{OH} = -10 \text{ mA}, V_{CC} = 3V$ | 4.2<br>2.3 |  |            | V |

Notes: 1. "Max" means the highest value where the pin is guaranteed to be read as low

2. "Min" means the lowest value where the pin is guaranteed to be read as high

3. Although each I/O port can sink more than the test conditions (20 mA at  $V_{CC} = 5V$ , 10 mA at  $V_{CC} = 3V$ ) under steady state conditions (non-transient), the following must be observed:

ATmega48PA/88PA/168PA/328P:

1] The sum of all  $I_{OL}$ , for ports C0 - C5, ADC7, ADC6 should not exceed 100 mA.

2] The sum of all  $I_{OL}$ , for ports B0 - B5, D5 - D7, XTAL1, XTAL2 should not exceed 100 mA.

3] The sum of all  $I_{OL}$ , for ports D0 - D4, RESET should not exceed 100 mA.

If  $I_{OL}$  exceeds the test condition,  $V_{OL}$  may exceed the related specification. Pins are not guaranteed to sink current greater than the listed test condition.

4. Although each I/O port can source more than the test conditions (20 mA at  $V_{CC} = 5V$ , 10 mA at  $V_{CC} = 3V$ ) under steady state conditions (non-transient), the following must be observed:

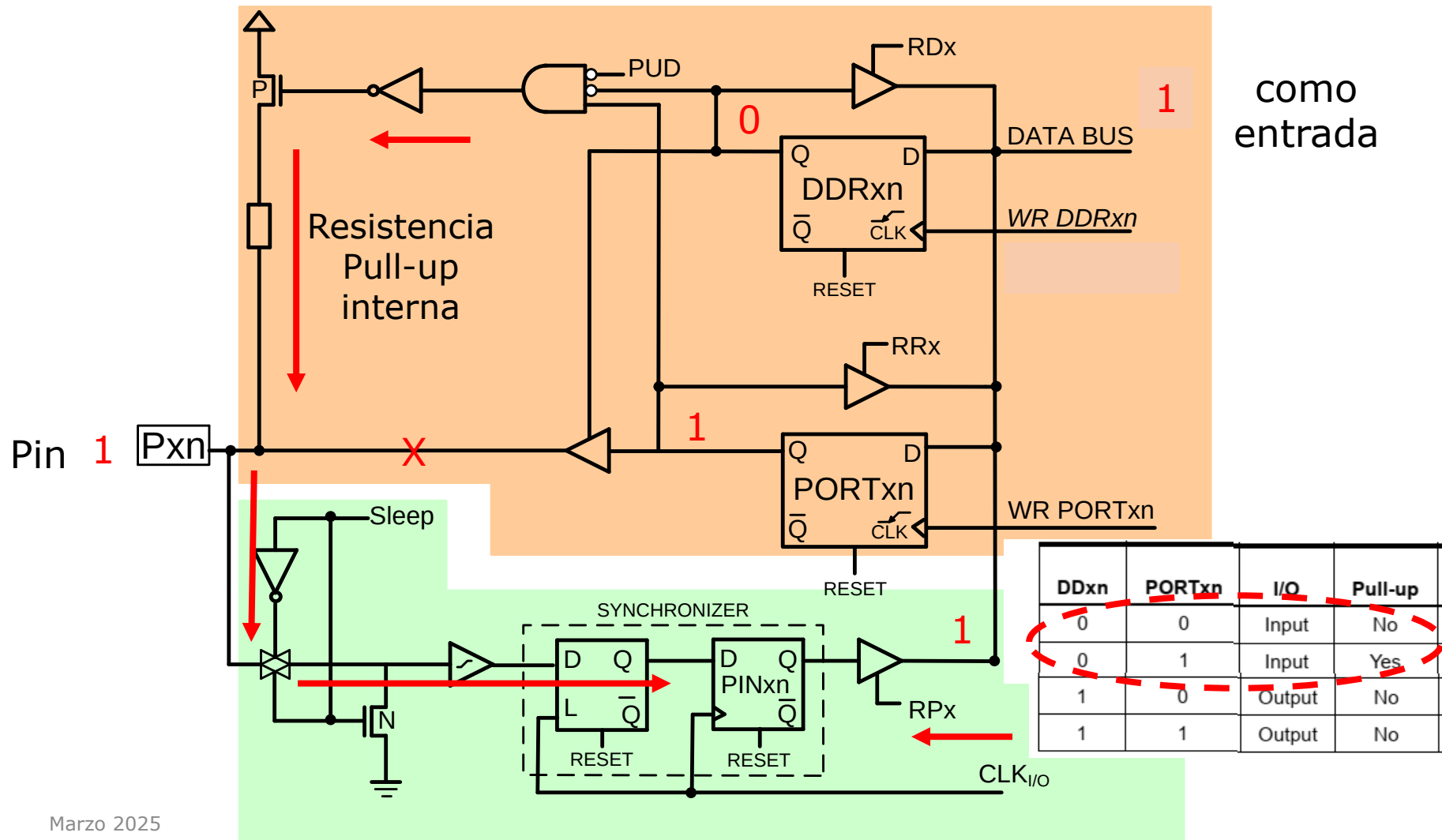
ATmega48PA/88PA/168PA/328P:

1] The sum of all  $I_{OH}$ , for ports C0 - C5, D0- D4, ADC7,  $\overline{\text{RESET}}$  should not exceed 150 mA.

2] The sum of all  $I_{OH}$ , for ports B0 - B5, D5 - D7, ADC6, XTAL1, XTAL2 should not exceed 150 mA.

If  $I_{OH}$  exceeds the test condition,  $V_{OH}$  may exceed the related specification. Pins are not guaranteed to source current greater than the listed test condition.

# Estructura de los Puertos I/O



# Características eléctricas de entradas

## 28.2 DC Characteristics

$T_A = -40^\circ\text{C}$  to  $85^\circ\text{C}$ ,  $V_{CC} = 1.8\text{V}$  to  $5.5\text{V}$  (unless otherwise noted)

| Symbol    | Parameter                                                           | Condition                                                                    | Min.                                   | Typ. | Max.                                   | Units            |
|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------|------|----------------------------------------|------------------|
| $V_{IL}$  | Input Low Voltage, except XTAL1 and $\overline{\text{RESET}}$ pin   | $V_{CC} = 1.8\text{V} - 2.4\text{V}$<br>$V_{CC} = 2.4\text{V} - 5.5\text{V}$ | -0.5<br>-0.5                           |      | $0.2V_{CC}^{(1)}$<br>$0.3V_{CC}^{(1)}$ | V                |
| $V_{IH}$  | Input High Voltage, except XTAL1 and $\overline{\text{RESET}}$ pins | $V_{CC} = 1.8\text{V} - 2.4\text{V}$<br>$V_{CC} = 2.4\text{V} - 5.5\text{V}$ | $0.7V_{CC}^{(2)}$<br>$0.6V_{CC}^{(2)}$ |      | $V_{CC} + 0.5$<br>$V_{CC} + 0.5$       | V                |
| $I_{IL}$  | Input Leakage Current I/O Pin                                       | $V_{CC} = 5.5\text{V}$ , pin low (absolute value)                            |                                        |      | 1                                      | $\mu\text{A}$    |
| $I_{IH}$  | Input Leakage Current I/O Pin                                       | $V_{CC} = 5.5\text{V}$ , pin high (absolute value)                           |                                        |      | 1                                      | $\mu\text{A}$    |
| $R_{RST}$ | Reset Pull-up Resistor                                              |                                                                              | 30                                     |      | 60                                     | $\text{k}\Omega$ |
| $R_{PU}$  | I/O Pin Pull-up Resistor                                            |                                                                              | 20                                     |      | 50                                     | $\text{k}\Omega$ |

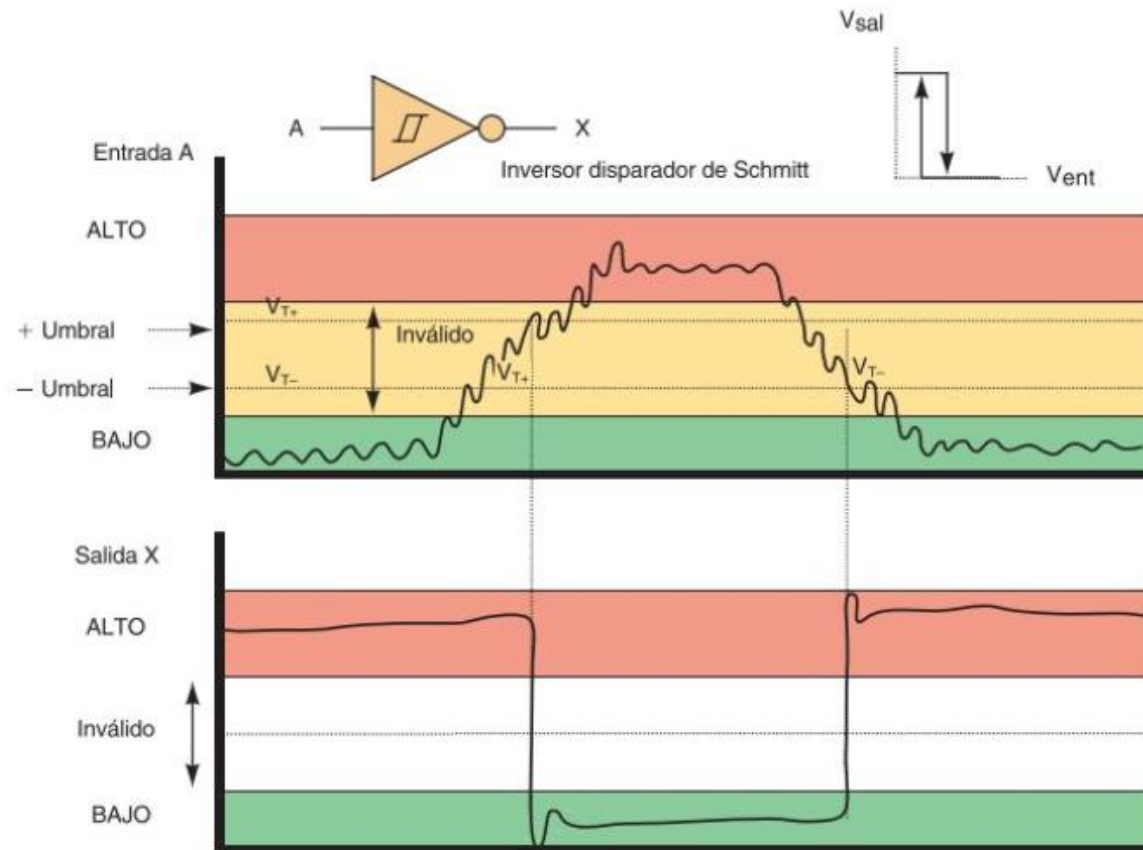
$V_{IL\text{max}} = 1.5\text{V} @ V_{CC} = 5\text{V}$

$V_{IH\text{min}} = 3\text{V} @ V_{CC} = 5\text{V}$



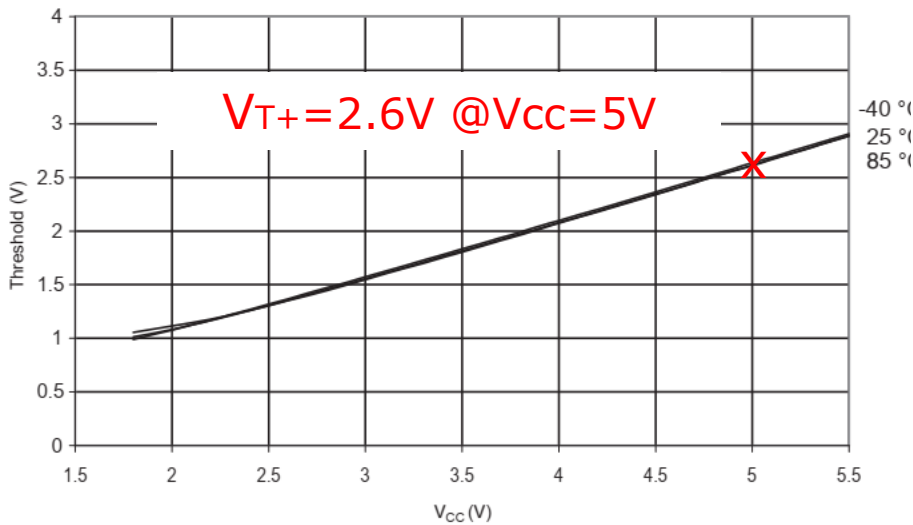
# Entrada Schmitt Trigger

Entrada Schmitt Trigger Permiten responder de manera confiable a entradas de variación lenta con ruido

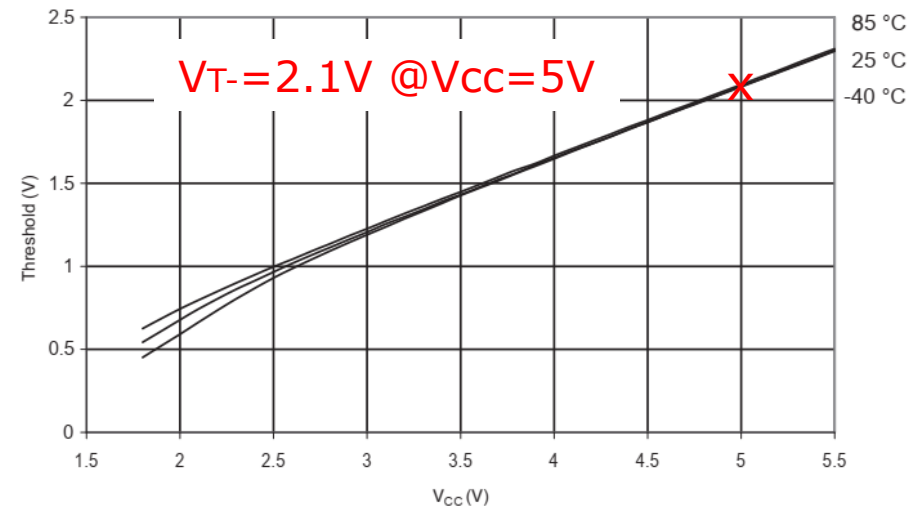


# Características eléctricas de entradas

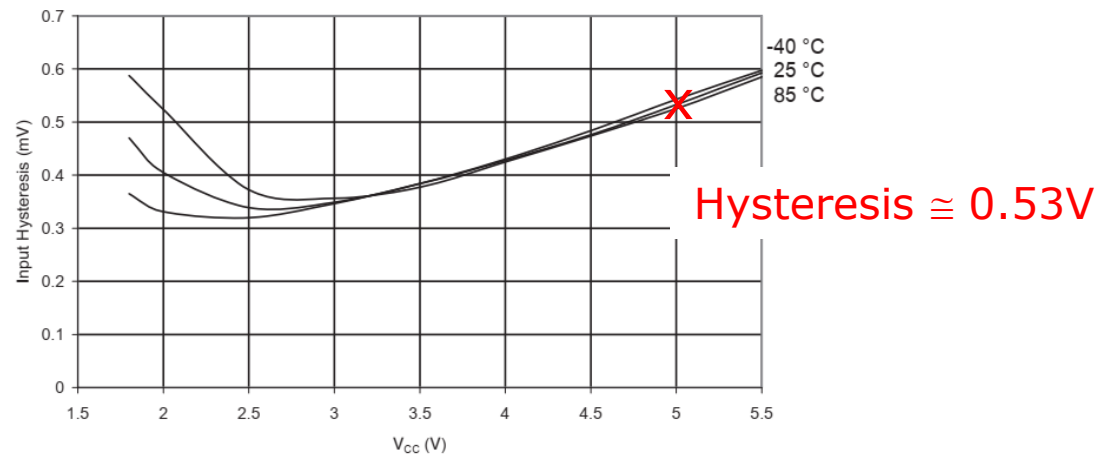
**Figure 29-163.** ATmega328P: I/O Pin Input Threshold Voltage vs.  $V_{CC}$  ( $V_{IH}$ , I/O Pin read as '1')



**Figure 29-164.** ATmega328P: I/O Pin Input Threshold Voltage vs.  $V_{CC}$  ( $V_{IL}$ , I/O Pin read as '0')

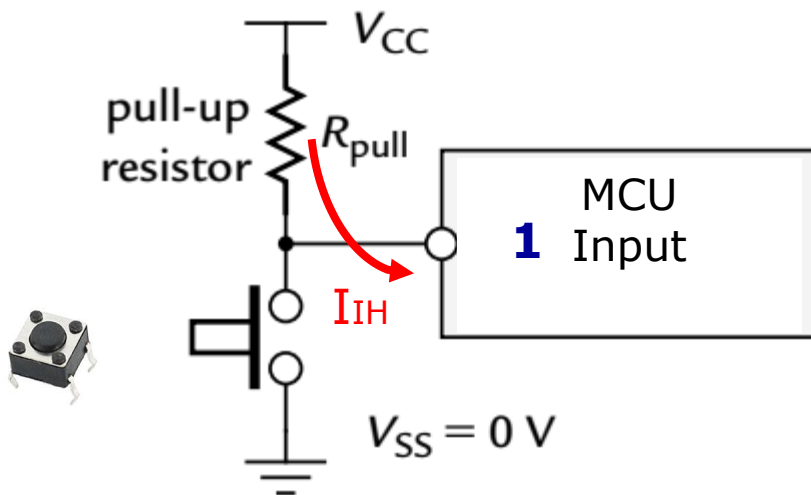


**Figure 29-165.** ATmega328P: I/O Pin Input Hysteresis vs.  $V_{CC}$



# Características eléctricas de entradas

- Utilizando un pulsador para establecer un valor lógico en una entrada



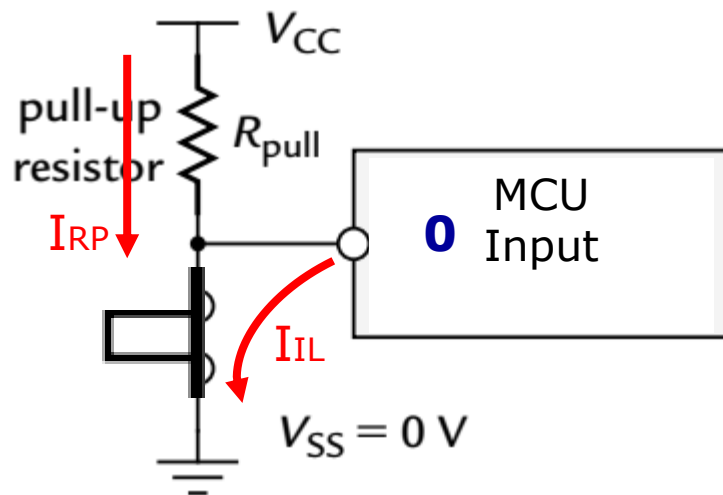
Pulsador no presionado: el estado lógico de la entrada es ALTO porque la misma se encuentra conectada a  $V_{CC}$  a través de  $R_{pull}$

de las especificaciones  $I_{IH} = 1\mu\text{A}$  (despreciable)

Por lo tanto a la entrada tengo  $V_{IH} \cong V_{CC}$

# Características eléctricas de entradas

- Utilizando un pulsador para establecer un valor lógico en una entrada



$$I_{IL} = 1\mu A \text{ (despreciable)}$$

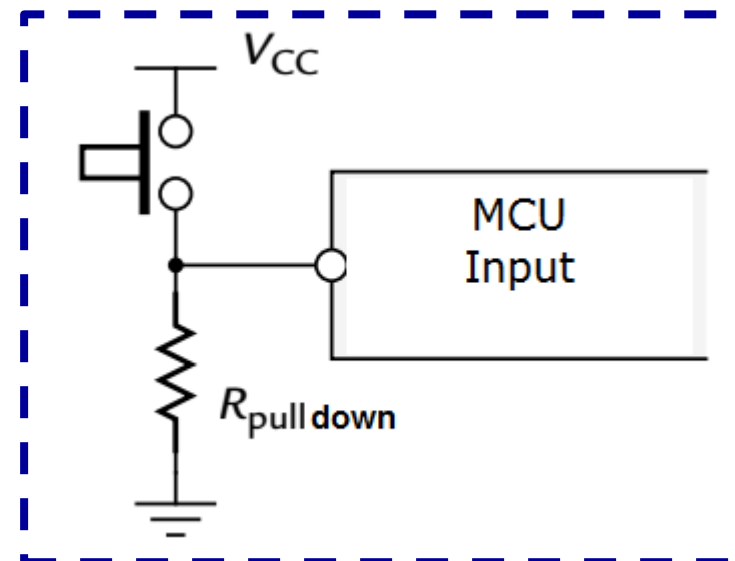
Por lo tanto a la entrada tengo  $V_{IL} \cong 0V$

$$I_{RP} = V_{CC} / R_{PU} \gg I_{IL}$$

Generalmente  $R_{PU} = 10K\Omega$

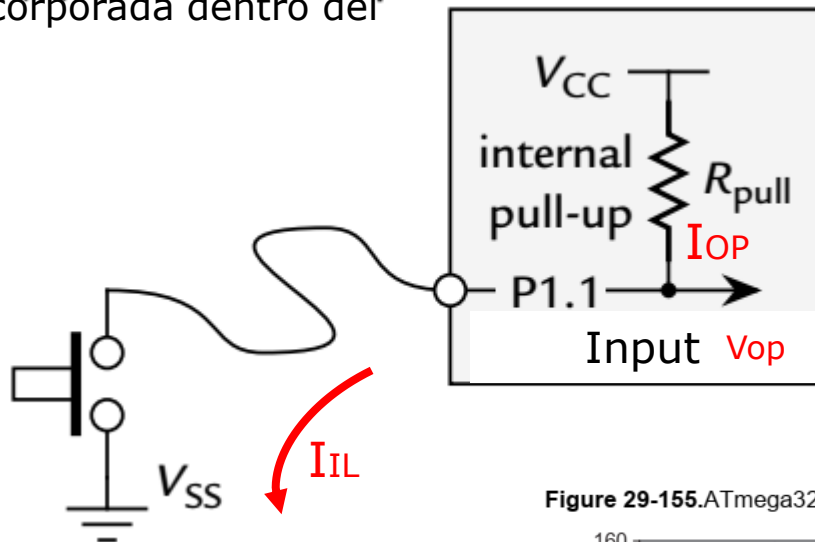
Pulsador presionado: el estado lógico de la entrada es BAJO porque la misma se encuentra conectada a 0V directamente

El mismo análisis puede realizarse utilizando una resistencia de Pull -Down



# Características eléctricas de entradas

Si usamos la resistencia de pull up que viene incorporada dentro del MCU :



Ahora ya sabemos como funcionan:

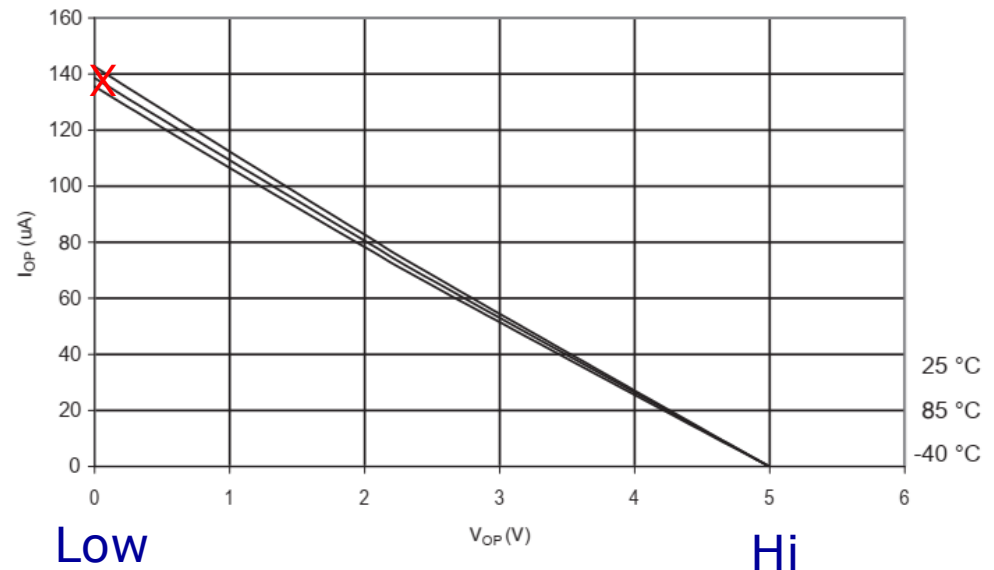
`pinMode(x,[INPUT,INPUT_PULLUP,OUTPUT])`

`digitalRead(x)`

`digitalWrite(x, [0,1])`

Para saber cuando vale  $I_{IL}$  cuando habilitamos el pull-up interno utilizamos la siguiente curva:

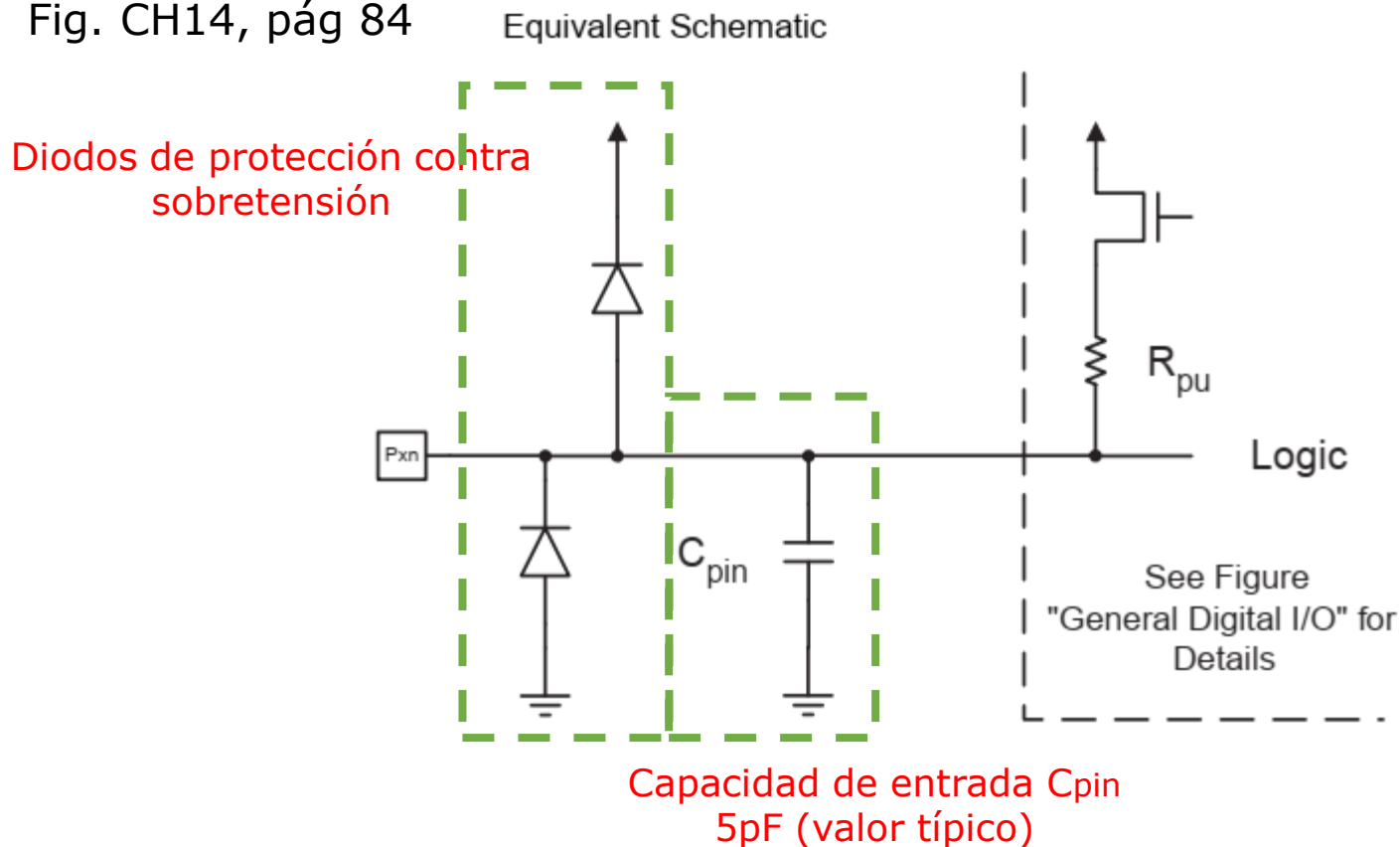
Figure 29-155. ATmega328P: I/O Pin Pull-up Resistor Current vs. Input Voltage ( $V_{CC} = 5\text{ V}$ )



# Características eléctricas de entradas

- Otras especificaciones que podemos encontrar en las hojas de datos:

Fig. CH14, pág 84



# Bibliografía:

- *The AVR microcontroller & Embedded Systems*. Mazidi, Naimis. (CH8 y **APENDICE C**)
- *Los Microcontroladores AVR de ATMEL*. Felipe Espinoza (CH3 y CH8)
- **Hoja de datos** ATMEGA328P (características eléctricas CH30)
- Extras:
  - **PLC open source con atmega AVR:**  
<https://www.electroallweb.com/index.php/2019/12/14/plc-con-arduino-atmega-328p-v4/>  
<https://www.controllino.com/>
  - **Entorno de programación PLC:**  
<https://docs.arduino.cc/software/plc-ide>